

$$\text{HA}2) \text{ a) } \frac{Y(s)}{R(s)} = \frac{\frac{2(s+1)}{s} \cdot \frac{1}{s}}{1 + \frac{s(s+1)}{s} \cdot \frac{1}{s} \cdot \frac{100}{s+100}} = \frac{(s+100)2(s+1)}{s^2(s+100) + s(s+1)100}$$

$$\text{b) } \frac{Y(s)}{W(s)} = \frac{s(s+100)}{s^2(s+100) + s(s+1)100}$$

$$\text{c) } E(s) = \frac{s^2(s+100)}{s^2(s+100) + s(s+1)100} R(s) \quad R(s) = \frac{1}{s}$$

$$\lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{s^2(s+100)}{s^2(s+100) + s(s+1)100} = 0$$

$$\text{d) } \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{s^2(s+100)}{s^2(s+100) + s(s+1)100} = \frac{100}{100} = 1 \quad \text{Ref } \frac{1}{s^2}$$

$$\text{HA}1) \quad G_S(s) = \frac{4}{1+2s} \quad G_R(s) = K_P \frac{1+sT_1}{sT_1} \quad D = \frac{1}{2} \sqrt{2} \quad \omega = 7$$

$$\cancel{G = G_S G_R = \frac{4}{1+2s} K_P \frac{1+sT_1}{sT_1} =}$$

$$G = \frac{G_S G_R}{1 + G_S G_R} = \frac{\frac{4K_P(1+sT_1)}{(1+2s)sT_1}}{1 + \frac{4K_P(1+sT_1)}{(1+2s)sT_1}} = \frac{4K_P(1+sT_1)}{(1+2s)sT_1 + 4K_P(1+sT_1)}$$

$$\cancel{T_1 = 0}$$

$$= \frac{4K_P(1+sT_1)}{2T_1s^2 + (T_1 + T_14K_P)s + 4K_P}$$

$$\Rightarrow s^2 + \left(\frac{1}{2} + 2K_P\right)s + \frac{2K_P}{T_1}$$

$$\frac{1}{2} + 2K_P = \frac{1}{2} \sqrt{2}$$

$$\frac{\frac{1}{2}\sqrt{2} - \frac{1}{2}}{T_1} = 7^2$$

$$2K_P = \frac{1}{2} \sqrt{2} - \frac{1}{2}$$

$$T_1 = \frac{1}{98} \sqrt{2} - \frac{1}{98}$$

$$K_P = \frac{1}{4} \sqrt{2} - \frac{1}{4}$$

Hausaufgabe 1 noch einmal
ordentlich gerechnet auf Seite 3

$$HA) s) a) \quad \frac{E(s)}{R(s)} = \frac{1}{1 + K \frac{s^2 + \alpha s + 1}{s^2 + \omega_0^2} \frac{1}{s(s+0,5)}} = \frac{s(s+0,5)(s^2 + \omega_0^2)}{s(s+0,5)(s^2 + \omega_0^2) + K(s^2 + \alpha s + 1)}$$

$$r(t) = \underset{0}{\sin}(\omega_0 t)$$

$$R(s) = \frac{\omega_0}{s^2 + \omega_0^2}$$

$$b) \quad \lim_{s \rightarrow 0} s E(s) R(s) = \lim_{s \rightarrow 0} \frac{s}{\omega_0^2 + s^2} \frac{s(s+0,5)(s^2 + \omega_0^2)}{s(s+0,5)(s^2 + \omega_0^2) + K(s^2 + \alpha s + 1)} = 0$$

$$c) \quad s(s+0,5)(s^2 + \omega_0^2) + K(s^2 + \alpha s + 1)$$

$$\omega_0 = 1, \alpha = 0,25 = \frac{1}{4}$$

$$s(s + \frac{1}{2})(s^2 + \frac{1}{16}) + K(s^2 + \frac{1}{4}s + 1)$$

nochmal schön gerechnet
siehe nächste Seite 3)

$$(s^2 + \frac{1}{2}s)(s^2 + \frac{1}{16}) + Ks^2 + \frac{1}{4}Ks + 1$$

$$s^4 + \frac{1}{2}s^3 + (\frac{1}{16}K + \frac{1}{16})s^2 + (\frac{1}{4}K + \frac{1}{32})s + 1$$

$$s^4 + \frac{1}{2}s^3 + (K+1)s^2 + (\frac{1}{4}K + \frac{1}{2})s + 1$$

$$a_3 a_2 a_1 - a_0 a_3^2 - a_1^2 a_4 > 0$$

$$\begin{aligned} d_0 &= 1 > 0 \checkmark \\ d_1 &= \frac{1}{2}K + \frac{1}{32} > 0 \text{ wenn } K > -\frac{1}{16} \checkmark \\ d_2 &= K + \frac{1}{16} > 0 \text{ wenn } K > -\frac{1}{16} \checkmark \\ d_3 &= \frac{1}{2} > 0 \checkmark \\ d_4 &= 1 > 0 \checkmark \end{aligned}$$

Hausaufgabe 1, Aufgabe c
noch einmal ordentlich
gerechnet auf Seite 3

$$\frac{1}{2}K^2 + \frac{3}{64}K + \frac{1}{32} - \frac{1}{4} - K^2 - \frac{1}{32} > 0$$

$$\frac{1}{2}(\frac{1}{4}K + \frac{1}{2})(K+1) - \frac{1}{4} - (\frac{1}{4}K + \frac{1}{2})^2 > 0$$

$$\frac{1}{8}K^2 + \frac{3}{8}K + \frac{1}{4} - \frac{1}{4} - \frac{1}{16}K^2 - \frac{2}{8}K - \frac{1}{4} > 0$$

$$+ K^2 + 2K - \frac{1}{4} > 0 \quad K_{1,2} = -1 \pm \sqrt{64}$$

$$-\frac{1}{2}K^2 - \frac{1}{64}K - \frac{1}{4} > 0$$

$$K^2 + \frac{1}{32}K + \frac{1}{2} < 0$$

$$K_{1,2} = -\frac{1}{64} \pm \sqrt{\frac{1}{64} - \frac{1}{2}} = -\frac{1}{64} \pm \sqrt{\frac{1}{2}} i$$

HA) 1)

$$G_R = K_p \frac{1+sT_1}{sT_1}$$

$$G_S = \frac{4}{1+2s}$$

$$D = \frac{1}{2} \sqrt{2}$$

$$\omega = 7$$

$$1 + G_R G_S = 0$$

$$1 + \frac{(K_p + K_p s T_1) 4}{(1+2s)sT_1} = 0$$

$$\frac{2s^2 T_1 + s T_1 + 4K_p + 4K_p s T_1}{-11-} = 0$$

$$\Rightarrow 2T_1 s^2 + (4K_p T_1 + T_1)s + 4K_p = 0$$

$$s^2 + (2K_p + \frac{1}{2})s + 2\frac{K_p}{T_1} = 0$$

$$\left. \begin{array}{l} s^2 + (2K_p + \frac{1}{2})s + 2\frac{K_p}{T_1} = 0 \\ s^2 + 2D\omega_0 s + \omega_0^2 = 0 \end{array} \right\} \begin{array}{l} \omega_0^2 = 49 = 2\frac{K_p}{T_1} \\ 2D\omega_0 = \sqrt{2} \cdot 7 = (2K_p + \frac{1}{2}) \end{array}$$

$$2K_p = \sqrt{2} \cdot 7 - \frac{1}{2}$$

$$T_1 = \frac{2}{49} K_p = 0,19$$

$$K_p = \frac{1}{4} (14\sqrt{2} - 1) = 4,7$$

HA) 3) c)

$$s(s+0,5)(s^2+\omega_0^2) + K(s^2 + \alpha s + 1)$$

$$(s^2 + 0,5s)(s^2 + \omega_0^2) + Ks^2 + \alpha sK + K$$

$$s^4 + 0,5s^3 + (\omega_0^2 + K)s^2 + (0,5\omega_0^2 + \alpha K)s + K$$

$$\omega_0 = 1$$

$$\alpha = 0,25$$

$$s^4 + 0,5s^3 + (1+K)s^2 + (0,5 + 0,25K)s + K$$

$$a_3 a_2 a_1 - a_0 a_3^2 - a_1^2 a_4 > 0$$

$$a_4 = 1 > 0 \checkmark$$

$$a_2 = (1+K)\frac{1}{4} > 0 \text{ wenn } K > -2$$

$$a_3 = \frac{1}{2} > 0 \checkmark$$

$$a_0 = K > 0 \text{ wenn } K > 0$$

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HA) 3)c)

$$a_0 = k$$

$$a_1 = \frac{1}{4}(2+k)$$

$$a_2 = 1+k$$

$$a_3 = \frac{1}{2}$$

$$a_4 = 1$$

$$a_3 a_2 a_1 - a_0 a_3^2 - a_1^2 a_4 > 0$$

$$\frac{1}{8}(2+k)(1+k) - \frac{1}{4}k - \frac{1}{16}(2+k)^2 > 0$$

$$\frac{1}{16}(2k^2 + 6k + 4 - 4k - k^2 - 4k - 4) > 0$$

$$k^2 - 2k > 0$$

$$k(k-2) > 0 \Rightarrow k-2 > 0$$

$$k > 2$$

~~$$k > 2 \text{ \& } k < 0 \downarrow$$~~

